



Question Bank FOR

Computer Networks

III Semester **BCA**

2025-2026
SEP Syllabus

**MANGALORE UNIVERSITY COMPUTER
SCIENCE TEACHERS' ASSOCIATION**

III SEMESTER BCA COMPUTER NETWORKS QUESTION BANK

UNIT-I

TWO MARKS

1. Define computer network.
2. Expand VPN, RFID.
3. Expand GPS, NFC.
4. Expand DMCA, CAPTCHA.
5. Expand WiMAX, NSP.
6. Expand ARPANET, ICMP.
7. Expand SMTP, RTP.
8. Expand HTTP, TELNET.
9. Expand FTP, DNS.
10. Expand TCP, UDP.
11. Expand DSL, SONET.
12. What is peer-to-peer model?
13. What is client-server model?
14. List the different forms of e-commerce.
15. What is profiling and phishing?
16. What are the two types of transmission technology?
17. What is unicasting and broadcasting?
18. What is communication subnet?
19. List and explain the components of subnet.
20. What are routing and forwarding algorithm?
21. What is addressing and flow control?
22. What is store-and-forward switching and cut-through switching?
23. List the different service primitives.
24. What is service and protocol?
25. Differentiate OSI and TCP/IP reference model.

THREE OR MORE MARKS

26. List and explain the uses of computer network.
27. Write a note on the following:

- i. Business application
 - ii. Home application
- 28. Write a note on the following:
 - i. Mobile users
 - ii. Social issues
- 29. Write a note on the following:
 - i. LAN
 - ii. MAN
- 30. Write a note on WAN.
- 31. Write a note on LAN.
- 32. Write a note on PAN.
- 33. Write a note on Internetworks.
- 34. Explain protocol hierarchies.
- 35. Explain design issues for the layers.
- 36. Differentiate connection oriented and connectionless service.
- 37. Write a note on service primitives.
- 38. Explain in brief different layers of OSI reference model.
- 39. Explain in brief different layers of TCP/IP reference model.
- 40. Differentiate OSI and TCP/IP reference model.

UNIT-II

TWO MARKS

- 1. Expand UTP, IrDA.
- 2. Expand STDM, CDMA.
- 3. Expand FDM, OFDM.
- 4. Expand TDM, CDM.
- 5. What is transmission medium?
- 6. What is meant by unguided media and guided media?
- 7. What are single-mode and multimode fiber.
- 8. Differentiate fiber optics and copper wire.
- 9. What is path loss?
- 10. What is multipath fading?
- 11. What is digital modulation?
- 12. What is multiplexing?

13. What is FDM and TDM?
14. List the different switching techniques.
15. Differentiate circuit switching and packet switching.
16. Write the different design issues of data link layer.
17. Write the various services offered by the data link layer to the network layer.
18. List the different framing methods.
19. Expand PPP, HDLC.
20. Expand NIC, FEC.
21. What are the two different approaches of Flow control.
22. List the different error-correcting codes.
23. Expand LDPC, CRC.
24. List the different fields from which the frame is composed of.
25. What is meant by **event=cksum_err** and **event=frame_arrival** ?

THREE OR MORE MARKS

26. Write a note on twisted-pair cable.
27. Write a note on the following: a) Coaxial cable b) Fiber optics
28. Write a note on radio transmission.
29. Write a note on microwave transmission.
30. Write a note on the following: a) Infrared transmission b) Light transmission
31. Explain the following types of multiplexing: a) FDM b) TDM
32. Explain Code Division Multiplexing.
33. Explain the different types of switching techniques.
34. Explain the various services provided by the data link layer to the network layer.
35. Explain the following framing methods: a) Flag bytes with byte stuffing
b) Flag bits with bit stuffing
36. Write a note on the following: a) Error control b) Flow control
37. Explain Hamming code of error-correcting codes.
38. Explain Binary convolutional code of error-correcting codes.
39. Explain Parity of error-detecting code with example.
40. Explain Checksums of error-detecting code with an example.
41. Suppose that a message **1001110010100011** is transmitted using Internet Checksum(4-bit word). What is the value of the checksum?
42. Explain Cyclic Redundancy Checks of error-detecting code with an example.
43. What is the remainder obtained by dividing x^7+x^5+1 by the generator polynomial x^3+1 ?

44. A bit stream **10011101** is transmitted using the standard CRC method. The generator polynomial is x^3+1 . Show the actual bit string transmitted. Suppose that the third bit from the left is inverted during transmission, Show that this error is detected at the receiver end.

UNIT – III

TWO MARKS

1. What is store-and-forward packet switching?
2. Write the various services offered by the network layer to the transport layer.
3. What is routing algorithm?
4. Expand MPLS , CIDR.
5. Compare between Virtual-Circuit networks and Datagram networks.
6. What is broadcast routing and multicast routing?
7. What is anycast routing?
8. What is Congestion?
9. List the approaches to congestion control.
10. List the different options for Option field in IPv4 protocol.
11. What is IP address? Give example.
12. What is meant by subnetting and subnets?
13. Expand NAT , SIPP.
14. List different special IP addresses.
15. What are the major goals of IPv6?
16. List different extension headers of IPv6.

THREE OR MORE MARKS

17. Explain the implementation of connectionless service.
18. Explain store and forward packet switching.
19. Explain the implementation of connection-oriented service.
20. Compare between Virtual-Circuit networks and Datagram networks.
21. Explain the concept of Flooding.
22. Explain Distance vector routing with an example.
23. Explain Link state routing with an example.
24. Explain hierarchical routing with an example.
25. Explain the approaches to congestion control.
26. List and explain the principles of network layer in the internet.

27. With a neat diagram explain the format of IP version 4 protocol.
28. Explain Classful addressing.
29. Briefly explain Network Address Translation.
30. With a neat diagram, explain the format of IP version 6 protocol.
31. Explain IPV6 extension headers.

UNIT – IV

TWO MARKS

1. Write down the various services offered by the transport layer.
2. Expand TSAP, NSAP.
3. List the two main protocols in transport layer.
4. Expand TCP , UDP.
5. What is error control?
6. What is flow control?
7. Draw a neat diagram of UDP header.
8. Expand RPC , RTP.
9. What is TCP?
10. Expand PAWS , SACK.
11. Expand DNS , ICANN.
12. How DNS is used?
13. List the different DNS resource record types.
14. What is Name resolution?
15. Expand SMTP, MIME.
16. What are the two kinds of subsystems of Email system.
17. What is meant by User agent?
18. List the different header fields related to message transport in RFC 5322.
19. List the different message headers added by MIME.
20. List different SMTP extensions.
21. Expand IMAP , SOAP.
22. List different IMAP commands.
23. What is Webmail?

THREE OR MORE MARKS

24. Explain Elements of Transport Protocol.
25. Explain the concept of Addressing.

26. Explain the process of connection establishment in Transport layer.
27. Explain the process of connection release in Transport layer.
28. With a neat diagram, explain User Datagram Protocol.
29. Explain the technique of Remote Procedure Call.
30. Explain Real Time Transport Protocol.
31. Explain TCP service model.
32. With a neat diagram, explain TCP segment header.
33. Explain the process of TCP connection establishment.
34. Explain the process of TCP connection release.
35. Explain the process of TCP sliding window.
36. Explain the architecture of Email system.
37. Explain the format of Internet message.
38. Write a note on MIME.
39. Write a note on SMTP.
40. Write a note on IMAP.
